Brown Invitational - Repaired practice edition

Division B | Rocks And Minerals | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

Name: _____ Date: _____

s2-10. (2 points) Authored replacement for source item s2-10: Supplied field notes place celestite with gypsum in evaporite beds. Describe an environment consistent with those observations.

Answer: _____
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s2-11. (1 points) Authored replacement for source item s2-11: Supplied crystallographic data: chalcopyrite is tetragonal and celestite is orthorhombic. Do these two specimens belong to the same crystal system?

Answer: _____
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s6-3. (2 points) Authored replacement for source item s6-3: Repair data: green amazonite is a variety of microcline; microcline is triclinic and orthoclase is monoclinic; both feldspars have composition KAlSi3O8. Explain which pair illustrates polymorphism.

Answer: _____
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_____

s6-5. (2 points) Authored replacement for source item s6-5: Can a green feldspar’s color or ultraviolet appearance alone establish that it contains lead? Describe a stronger test.

Answer: _____
_____
_____

s15-1. (4 points) Authored replacement for source item s15-1: Label these supplied descriptions: A is a sheet parallel to bedding; B cuts across bedding; C is a concordant dome-shaped intrusion; D is an intrusive body with exposed area greater than 100 square kilometres.

Answer: _____
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_____

s16-1. (2 points) Authored replacement for source item s16-1: Name and explain these four relative-dating principles: superposition, original horizontality, lateral continuity and cross-cutting relationships.

Answer: _____
_____
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s17-1. (3 points) Authored replacement for source item s17-1: Replace the missing facies diagram using this supplied schematic: A has moderate pressure and lower temperature; B has similar pressure and higher temperature; C has high pressure and low temperature. The supplied candidate facies are greenschist, amphibolite and blueschist. Match each once.

Answer: _____
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_____

s17-2. (3 points) Authored replacement for source item s17-2: Explain why hornfels, greenschist, blueschist, amphibolite and granulite cannot all be put into one universal lowest-to-highest temperature grade sequence without pressure information.

Answer: _____
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